Apriori and FP-growth on Sparse Transactions: Computational Cost and Association Rule Stability

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***Abstract*—Lowering minimum support uncovers combinations that occur in few baskets. It also expands the search and makes the confidence of individual rules sensitive to a few observations. We implemented Apriori and FP-growth in Python and compared their outputs and running times on 1892 transactions containing 1990 distinct items. Both algorithms returned identical itemsets and support counts at four thresholds. Reducing minimum support from 2% to 0.2% increased the number of frequent itemsets from 96 to 6,524 and widened the observed runtime ratio from 1.57 to 17.39 in favour of FP-growth. Apriori spent most of its time processing pairs and triples at the lowest threshold. At 1% support and 50% confidence, 49 rules qualified. For 27 of them, the fifth percentile of bootstrap confidence still met the cutoff. Two rules with identical lift but different antecedents illustrate how this criterion distinguishes the more consistent direction.**

***Index Terms*—Apriori, association rules, bootstrap, FP-growth, frequent itemsets.**

I. Introduction

In a sparse transaction database, two items can occur together in nearly every basket containing one of them and still appear in only a small fraction of all baskets. A high support threshold can discard such a pair. Lowering the threshold poses two questions. How much additional work does it take to recover the rarer combinations? Among the rules they yield, which consistently meet the chosen confidence threshold?

Apriori [1] and FP-growth [2] provide a controlled comparison for the first question. Apriori counts explicit candidates at successive itemset lengths. FP-growth stores shared transaction prefixes in a tree and mines conditional patterns. Both seek the same frequent itemsets, so agreement on every support count lets us compare the cost of two search strategies on identical output.

We implemented both miners and a common rule generator, verified their results against a worked example and randomized tests, and recorded per-level candidate counts and elapsed times. For the second question, we evaluated twelve combinations of four support and three confidence thresholds. We then resampled whole transactions to examine how confidence varied for each rule at a selected operating point.

II. Data and Methods

The course database contains 1892 baskets of integer item identifiers (Table I). A basket contains 11.79 items on average. Only 279 of the 1990 identifiers appear in at least 1% of baskets. We therefore swept minimum support from 2% down to 0.2%. For detailed rule analysis we used 1%, which requires at least 19 baskets per itemset and yields a rule set small enough to inspect in full.

TABLE I
Basic statistics of the transaction database.

Quantity	Value
Transactions	1892
Distinct items	1990
Total item occurrences	22,300
Mean transaction length	11.79
Shortest / longest transaction	5 / 86
Items covering $\geq 5\%$ of transactions	14
Items covering $\geq 1\%$ of transactions	279

A. Mining the Same Itemsets by Two Routes

For a database of n baskets, let $\sigma(S)$ count the baskets containing every item in S. An itemset is frequent if $\sigma(S) \geq \lceil ns \rceil$ for minimum support s. The four thresholds in descending order require 38, 19, 10 and 4 baskets, respectively.

Apriori represents each itemset as a sorted tuple. To form a candidate of length k, it joins two frequent $(k-1)$-itemsets with a common prefix of length $k-2$ and checks that every immediate subset is frequent. Each surviving candidate enters the support-counting stage. The implementation estimates the number of basket subsets, $\sum_t \binom{|t|}{k}$, and the number of candidate–basket comparisons, $n|C_k|$, where C_k denotes the candidate set after the subset check. It chooses the counting strategy with the smaller estimate. For each level we record $|C_k|$, the number meeting minimum support, and the time for generation and counting together.

FP-growth counts individual items, drops infrequent ones and inserts the remaining items in descending frequency order into a prefix tree. Baskets with a common prefix share nodes. For each suffix item, the miner collects its prefix paths, weights them by their counts and recursively mines the resulting conditional tree. Both miners return itemsets and integer support counts in the same representation.

B. Implementation Checks and Timing

We first reproduced the four-basket course example [3]. At 50% support, six candidate pairs yield four frequent pairs. Only one triple, $\{2,3,5\}$, survives. The example also checks rule direction. Both baskets containing item 1 contain item 3, whereas only two of the three baskets containing item 3 contain item 1.

We compared Apriori against exhaustive enumeration on 30 randomized databases at four thresholds each. A second

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TABLE II
EFFECT OF THE SUPPORT THRESHOLD ON SCALE AND COST FOR THE TWO MINERS.

min_sup	Itemsets	$k_{\max}$	Apriori (s)	FP-growth (s)
2%	96	2	0.022	0.014
1%	335	3	0.098	0.035
0.5%	1,115	4	0.509	0.097
0.2%	6,524	7	3.999	0.230

test compared FP-growth with Apriori on 25 randomized databases at four thresholds each. The 23 automated tests also cover subset pruning and threshold boundaries. On the course database, we compared the complete itemset-to-count mappings at all four support thresholds.

The single-threaded Python 3.13.14 benchmark timed the mining calls on the same loaded baskets. If the first call took less than 0.8 s, the driver reported the minimum of three runs. Otherwise it retained the first time. Per-level measurements come from the first Apriori call. Rule generation and file export were outside the timers.

C. Rule Generation and Resampling

For every frequent itemset with at least two items, we enumerate each nonempty proper subset Y as an antecedent and use the remaining items X as its consequent. Writing $p(S) = \sigma(S)/n$, the two measures for $Y \to X$ are

$$\mathrm{conf}(Y \to X) = p(X \cup Y)/p(Y), \tag{1}$$

$$\mathrm{lift}(Y \to X) = \mathrm{conf}(Y \to X)/p(X). \tag{2}$$

We retain rules at confidence thresholds of 50%, 60% and 70%. At the working setting of 1% support and 50% confidence, we hold the selected rules fixed. Each of 200 bootstrap replicates draws n whole baskets with replacement and recomputes the measures. The 5th and 95th percentiles form a central 90% bootstrap interval. Our stability criterion asks whether the lower confidence endpoint reaches 50%.

III. Results and Discussion

A. Lower Support Widens the Runtime Gap

Table II compares elapsed time for identical outputs. As minimum support fell from 2% to 0.2%, the miners recovered 6,524 itemsets instead of 96, including patterns as long as 7 items. FP-growth took less time at each threshold. At 2%, the difference was eight milliseconds. At 0.2%, Apriori took 3.999 s and FP-growth took 0.230 s, a 17.39-fold difference.

Fig. 1 shows which pattern lengths account for the larger output. At 1% support only four triples qualify. At 0.2%, the output includes 1,255 triples and 324 four-item sets, alongside a single seven-item set. Lower support therefore expands the search across both pairs and longer patterns.

Table III identifies the expensive levels. At 0.2% support, 1,517 frequent singletons gave Apriori 1,149,886 possible pairs. Every pair of frequent singletons passes the subset check, yet only 3,356 pairs met the support threshold. Across levels, 6,524 of 1,161,310 counted candidates were frequent

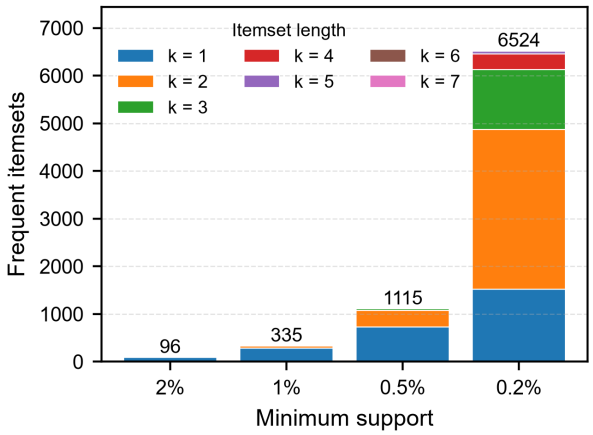


Fig. 1. Frequent itemsets by length. Each bar represents one support threshold, with the total shown above it.

TABLE III
PER-LEVEL CANDIDATES, RETAINED ITEMSETS AND TIME FOR THE WORKING THRESHOLD B (1%) AND THE STRESS THRESHOLD D (0.2%).

Level	k	Candidates	Retained	Time (s)
B	1	1,990	279	0.004
B	2	38,781	52	0.085
B	3	14	4	0.009
D	1	1,990	1,517	0.005
D	2	1,149,886	3,356	2.057
D	3	8,891	1,255	1.650
D	4	471	324	0.263
D	5	62	62	0.018
D	6	9	9	0.003
D	7	1	1	0.002

(0.56%). This percentage measures retention after support counting.

The pair and triple levels took 2.057 and 1.650 s, respectively, or about 93% of the recorded level time together. The triple level took nearly as long as the pair level despite admitting far fewer candidates to counting. FP-growth represented the filtered baskets in a main tree with 18,851 nodes at this threshold and mined extensions through conditional trees.

B. Support Controls Which Rules Enter the Analysis

Table IV separates the effects of support and confidence. At 50% confidence, reducing support from 2% to 1% raised the rule count from 5 to 49. At 0.2% support, 6,068 rules qualified, and 2,959 still qualified at 70% confidence. At the 2% support threshold, no rule reached 70% confidence.

The most frequent item appeared in 189 of 1892 baskets. By subset containment, every nonempty consequent has support at most $189/1892$. A rule with confidence at least 0.50 therefore has lift at least $0.50/(189/1892) \approx 5.0$. This bound explains why the lift-above-one filter retains every rule in Table IV.

At 0.2% support, one rule attaining the maximum lift of 473 was $\{30, 1546\} \to \{1430, 1541\}$. Its antecedent and consequent each occurred in the same four baskets. Confidence was therefore one, and lift was $1/(4/1892) = 473$. Reporting the four baskets alongside the ratio makes its scale interpretable.

TABLE IV
NUMBER OF ASSOCIATION RULES BY PARAMETER SETTING. EVERY RULE IN THIS TABLE SATISFIES lift > 1.

min_sup	50%	60%	70%
2%	5	2	0
1%	49	29	15
0.5%	192	117	71
0.2%	6,068	4,346	2,959

C. Rule Direction Changes the Stability Decision

At the working setting, joint support ranged from 19 to 78 baskets, with a median of 25. Among the 49 selected rules, 27 (55.1%) retained a lower confidence endpoint of at least 50%. Table V gives both measures for the eight highest-lift rules.

Fig. 2 shows the spread of lift estimates among the twelve highest-ranked rules. The two directions of the leading pair have the same interval, a consequence of the symmetry of lift. Their confidence estimates reveal a distinction that the lift plot conceals.

Items 86 and 27 shared 23 baskets. Of the 25 baskets containing item 86, 23 also contained item 27, giving confidence 0.92. Reversing the rule changed the denominator to 42 and reduced confidence to 0.55. Both directions had lift 41.44 and the same bootstrap lift interval $[33.78, 55.65]$. Their confidence intervals were $[0.84, 1.00]$ and $[0.43, 0.68]$, respectively. Both directions met the cutoff in the observed baskets, but only $86 \to 27$ retained it at the lower bootstrap endpoint. The antecedent count and confidence interval thus distinguish two rules that lift ranks equally.

IV. CONCLUSION AND FUTURE WORK

On 1892 sparse transactions, Apriori and FP-growth agreed on every frequent itemset and support count. FP-growth's observed runtime advantage grew from 1.57-fold to 17.39-fold as support fell. Apriori's level timings locate most of the work at pairs and triples. The rule analysis shows what the extra search yields: thousands of low-support rules, some with high lift based on just four baskets. At the 1% working threshold, bootstrap confidence narrowed 49 rules to 27 that retained the directional confidence threshold at the lower endpoint.

A follow-up benchmark could vary basket density and repeat each timing under a common protocol, recording candidate generation, counting and peak memory separately. A labelled transaction dataset with timestamps would also allow the rules selected by resampling to be evaluated on later baskets and interpreted in terms of the items involved.

TABLE V
STRONGEST RULES AT THE WORKING THRESHOLD WITH 90% BOOTSTRAP INTERVALS ($B = 200$).

Y	X	supp	conf	90% CI	lift	90% CI
{86}	{27}	0.0122	0.92	[0.84, 1.00]	41.4	[33.8, 55.6]
{27}	{86}	0.0122	0.55	[0.43, 0.68]	41.4	[33.8, 55.6]
{185}	{57}	0.0111	0.70	[0.54, 0.82]	34.0	[25.8, 44.7]
{57}	{185}	0.0111	0.54	[0.39, 0.68]	34.0	[25.8, 44.7]
{44}	{55}	0.0153	0.63	[0.50, 0.76]	29.8	[24.1, 38.2]
{55}	{44}	0.0153	0.72	[0.61, 0.83]	29.8	[24.1, 38.2]
{79}	{17}	0.0122	0.57	[0.46, 0.69]	25.9	[20.7, 33.5]
{17}	{79}	0.0122	0.55	[0.41, 0.67]	25.9	[20.7, 33.5]

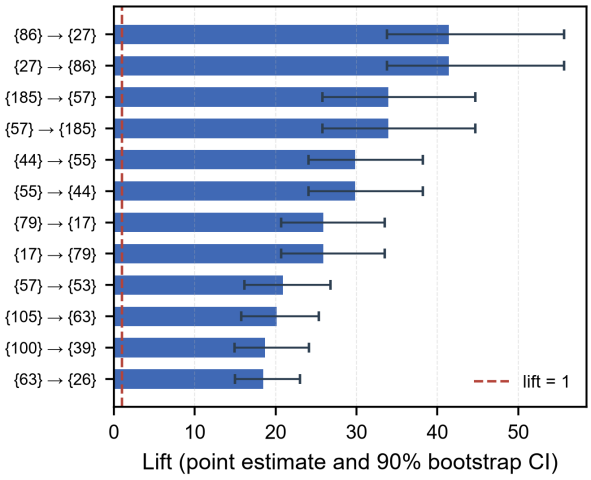


Fig. 2. Lift and central 90% bootstrap intervals for the twelve highest-lift rules at 1% support and 50% confidence.

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